PERCEIVING CORRELATIONS FROM SCATTERPLOTS

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This study investigated the effects of statistical training and various perceptual characteristics of scatterplot displays on intuitive estimates of correlations. University professors and first year undergraduate students estimated correlations from scatterplots (1) with three different levels of correlation in the data, (2) with or without a regression line, and (3) with three different types of dispersion of the data point clouds. The faculty generally used higher and a wider range of values, but both groups perceived a higher correlation when a regression line was present and both groups were equally influenced by the different types of dispersion of the point cloud. These findings indicate that the estimation of correlations from scatterplots is a perceptually based process, which is largely independent from formal statistical training.

INTRODUCTION

The visual inspection of the bivariate distribution of data points is one of the best ways to gain insights into the pattern and peculiarities of a given set of data. As such, the viewing of a scatterplot may be an important part of the evaluation of data during the course of a decision making process. But correlation estimates from scatterplots vary systematically from accepted statistical measures of correlation. Relative to the statistical measures used to indicate the strength of the correlation, people typically underestimate correlations from scatterplots (e.g., Jennings, Amiable, and Ross, 1980; Cleveland, Diaconis, and McGill, 1982). In addition, estimates based on scatterplots are affected by various display characteristics, such as the dispersion of the point cloud (Cleveland, Diaconis & McGill, 1982; Lauer & Post, 1989).

Understanding these phenomena has some applied value. Thus, Bobko and Karren (1979) have shown how perceptual attributes of the display and cognitive processes involved in estimating correlations from them may bias legal decisions. With the fast spreading use of powerful graphic software packages for personal and business computers and the incorporation of (or ability to incorporate) scatter plots in many decision support systems, the cognitive aspects of the assessment of correlations from them have gained further applied importance.

In this study we investigated the impact of various statistical and perceptual properties of scatter plots on estimates of correlation. Some of these characteristics can easily be manipulated by the graphic designer who draws the scatter plot, thereby manipulating the impressions generated by the displayed data.

We were specifically interested in three display variables: the strength of the statistical association, the shape of the point cloud, and the impact of a regression line drawn through the data. We attempted to determine the effect these variables have on people with different levels of statistical training: people with extensive statistical training (university faculty) and people with limited knowledge in statistics (first year undergraduate students).

Three different levels of correlation, considered moderate to high, and three different types of dispersion of the data-point clouds were presented. The product-moment coefficient (r) assumes bi-variate normal distributions, which generally cause oval dispersions of the data-point cloud. Two additional types of point clouds were tested in the present experiment. One had a uniform dispersion of points around the regression line, while the other had a trumpet-shaped type of dispersion, in which the variance of the data-points increases at higher levels of X. This is similar to the case of outliers and we expected to find relative higher estimates of correlation for this type of dispersion than for the other two types. The trumpet-shape was selected in order to retain the linearity of the least-square prediction line, which might be lost if outliers are positioned randomly in space. Finally, we tested the impact of including regression lines on correlation estimates. We hypothesized that the line would act as a perceptual center of gravity that would diminish the perceptual effects of the scatter of the data point cloud, and thus increase the estimated correlation.

METHOD

Subjects

Two groups of subjects participated in this experiment: 'novices' consisting of 19 first- and second-year undergraduate students in the Department for Behavioral Sciences, who had completed only a basic introductory course in statistics; and 'experts' consisting of 10 faculty members involved in quantitative research.

Apparatus, Design, and Procedure

Each subject received a booklet containing 54 different scatterplots, each with 21 data points. The variable on the X-axis had seven discrete levels, which were labeled consecutively from 100 to 700. At each level of this variable there were three data points in the space. A computer program generated the data points, so that a predetermined correlation (accurate at +.01) and a predetermined shape of the data point cloud were obtained. Two of the three points were generated randomly and the third point was computed, so that the average of the three data points for a given value of X would fall on the linear regression line. Thereby we assured that the best possible correlation coefficient is the linear one, and no nonlinear relations were present.

The 54 stimuli presented to a subject were made out of the following combinations: (a) three repetitions of (b) three levels of correlation (r^2=.40, .65 and .90), (c) with three types of dispersion of the data point clouds (uniform around the regression line; oval, with decreasing variance towards the lower and higher levels of X; and trumpet-shaped, with increasing variance at the higher levels of X), and (d) with or without the regression line. Each set of data that made up a display was presented to each subject only once. The order of the displays was randomized within and between subjects. Subjects indicated their correlation estimate by marking a scale printed under the scatterplot, ranging from 0 to 100 with tick marks for every 10 units. Subjects' task was to check the degree of correlation which seemed appropriate, considering that 0 means no relation and 100 means a perfect correlation. They worked individually, were not constrained in time, and required approximately 15 minutes to complete the experiment.

RESULTS

A four-way ANOVA, of Group (faculty vs. students), Correlation Level (r^2=.4, .65, and .9), Dispersion of the data point cloud (uniform, oval, and trumpet shaped), and existence or absence of the regression line, yielded significant main effects for all variables, and three significant two-way interactions. The greatest and most significant effect was that of the correlation level [$F(2,54)=112.68$, $p<.0001$]. The subjects' estimates for the squared correlation levels of .40, .65 and .90 were .335, .475, and .670, respectively. These estimates fall considerably beneath the true values of r^2 (and even more so beneath the corresponding values of r which are .63, .81 and .95). The faculty gave generally higher estimates than students [r(est)=.57 vs. .45; $F(1,27)=8.71$, $p<.01$], and the magnitude of the difference was larger for the higher correlation levels. However, the faculty too "underestimated" the correlation relative to the objective values of r^2. These effects, as well as the significant interaction Group X Correlation Level [$F(2,54)=3.95$, $p<.03$], are presented in Figure 1.

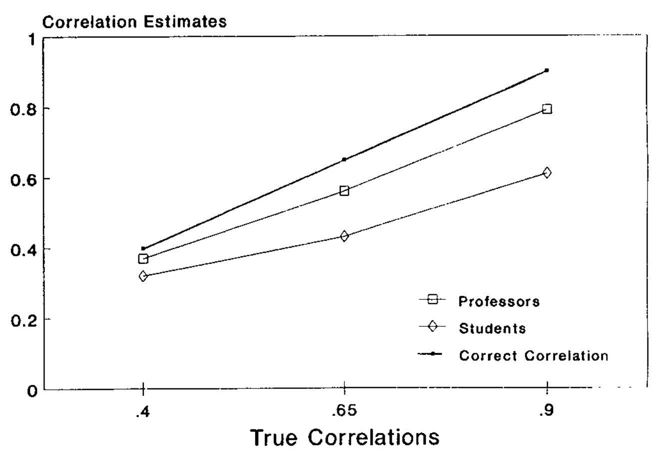

Figure 1. Correlation estimates as a function of the objective correlation level (r^2) and the statistical training (university faculty vs. undergraduate students).

The dispersion of the point clouds interacted with the level of correlation [$F(4,108)=2.79$, $p<.03$]. The effects are displayed in Figure 2, from which it can be seen that across all levels of correlation, estimates for the trumpet shaped type of dispersion were the highest [r(est)=.53] and estimates for the oval cloud were the lowest [r(est)=.47]. Furthermore the interaction indicates that for the low level of correlation the major difference among the three shapes is between the trumpet-shaped cloud and the others, while for the high level of correlation the major difference is between the oval cloud and the others.

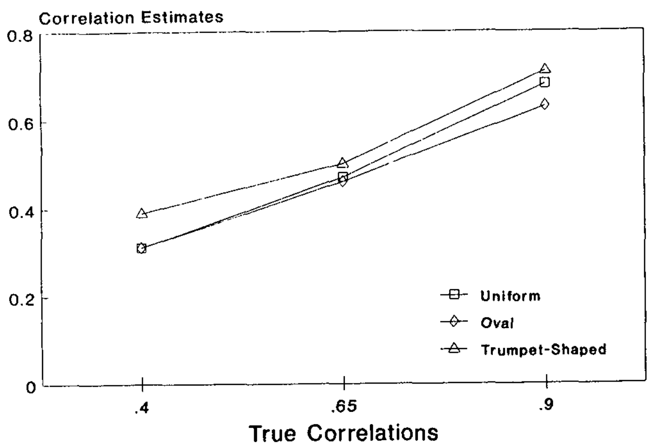

Figure 2. Correlation estimates as a function of the objective correlation level (r^2) and the type of dispersion of the point cloud.

Displaying the regression line caused an increase in the estimates (and brought them closer to r^2) relative to the estimates based on the same scatterplots without a regression line [r(est)=.53 vs. r(est)=.46; $F(1,27)=19.12$, $p<.0005$]. Another effect of the regression line - observable in Figure 3 - was to minimize differences in the estimates due to the types of dispersion [$F(2,54)=3.96$, $p<.025$, for the interaction Dispersion X Line].

An important finding was the fact that statistical experience did not interact with the dispersion of the point cloud or the presence of the regression line. Thus it seems that subjects' statistical training did not moderate the effects that these perceptual factors had on the perceived correlations.

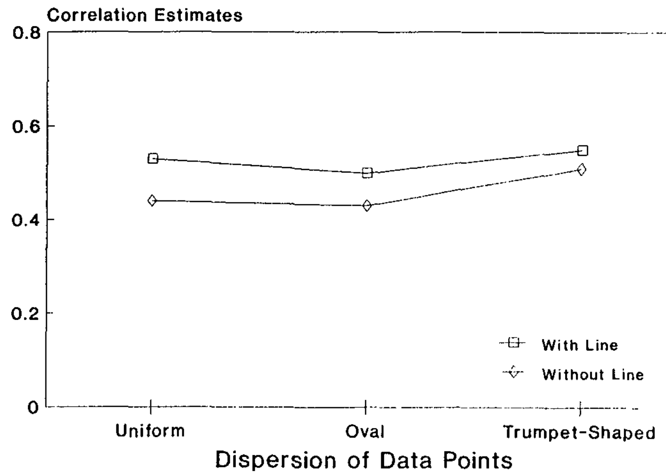

Figure 3. Correlation estimates for the three types of dispersion of the point clouds, with and without the regression line drawn in.

DISCUSSION

The present study demonstrated that subjects consistently underestimate correlations relative to the objective measures r and r^2. These underestimations are present both for people with and without statistical training, although knowledge in statistics leads to slightly higher estimates. This finding itself is not surprising since the Pearson Product Moment Coefficient - the most ubiquitous measure of linear correlations between variables - is an arbitrary measure of correlation and any monotonic transformation of its value might serve equally well as coefficient of correlation.

The important question is, why no function has yet been found, which can appropriately describe the relation between correlation estimates and objective statistical measures. Our findings suggest that previous attempts have failed because they focused on strictly quantitative aspects of the data - with the total exclusion of the perceptual factors inherent in the different ways the data have been displayed. The interaction between the level of correlation and the type of dispersion of the data-point clouds indicates that any general function that will relate estimates and objective measures, cannot ignore a number of graphical properties of the display.

One graphical feature that tended especially to enhance estimates was the presence of a regression line. When this line appears in a scatterplot, subjects' estimates are higher and they are less influenced by deviations from the bi-variate normal distribution of data-points.

Both subjects with and without statistical training were equally influenced by these perceptual characteristics of the display. It therefore seems that formal statistical training does not make decision-makers less susceptible to the influence of perceptual properties of the displays they see. Thus, estimating correlations from scatterplots seems to involve mechanisms which are different from those applied in usual statistical tasks.

Further inquiries into the impact of various graphic features on the interpretation of displays may have definite theoretical and applied value. Additional research is now underway to determine the effects of other variables on the perceived correlations.

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